
XN Project 1 - Brain Hemorrhage CT Image Segmentation and Classification

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Abstract

1 A brain hemorrhage and more specifically, an intracranial hemorrhage, is a medical
2 condition in which bleeding occurs within the skull, specifically inside the brain
3 tissue or in the space between the brain and the skull. It can be caused by trauma,
4 high blood pressure, aneurysms, blood clotting disorders, or other underlying
5 medical conditions. Intracranial hemorrhages can be life-threatening and require
6 prompt medical attention to prevent further damage to the brain. CT images of
7 segmentation of hemorrhage is a crucial step in the diagnostic process since it
8 helps doctors to locate and measure the bleeding accurately. The research in brain
9 CT image hemorrhage segmentation has a practical application in the medical
10 field. The research can develop accurate and efficient methods, which can help
11 the doctors in tracking the process of the patient's conditions. Additionally, the
12 identification of hemorrhage in CT images is important in the treatment of the
13 patients. If a hemorrhage is timely and accurately detected, it can help doctors make
14 the right decisions of surgical interventions or administration of medications. This
15 is especially crucial since intracranial hemorrhages come in four main subtypes
16 based on where the bleeding takes place between the skull and brain (dura).

17 1 Introduction

18 1.1 Data sources

19 Our image data is from Zeta Surgical, a biotechnology company located in Boston. They provided
20 752794 layered (post-sliced) CT scans of brain hemorrhages in 512x512 images for 7 types of
21 intracranial hemorrhages. For each hemorrhage, there were four images taken at different angles as
22 well as different "windows" which included the brain and bone window. Windows are important for
23 visualizing head-level CT scans with bone windows being better at visualizing features of the skull
24 while brain windows parse out the granularity within tissue better.[4] We also obtained .csv files for
25 each type of intracranial hemorrhage: subdural, epidural, subarachnoid, and intraparenchymal. Each
26 .csv file contained labeled coordinates which formed polygons which designated the location of the
27 hemorrhage on the CT scan. Each entry corresponded to one hemorrhage and four images so they
28 were pre-labeled with a case ID. There were further data corresponding to each entry such as the
29 amount of hemorrhages (regions of interest), the number of qualified labelers of the hemorrhage, the
30 agreement among the labelers, as well as comments.

1.2 Related work

There have been multiple prominent implementations of machine learning models for the detection of brain hemorrhaging. Some of the most famous include: 1) An image segmentation model built for non-contrast CT images developed by researchers in the Netherlands[5]. This study used a 3D convolutional neural net to segment the images. The model was successful with a dice coefficient of .91 2) Implementations of novel deep learning techniques to fully convolutional networks as well as other artificial neural networks [6][7]

2 Analysis

2.1 Methodology

The data we received required some pre-processing of the images such as removing corrupt or flagged images from the dataset as they were mislabeled or unreadable. After removing corrupt data and attaching the proper labels with their corresponding images, we began the implementation of our machine learning models. We used a logistical regression model, a convolutional neural network, support vector machine model, and implemented image segmentation to determine the best method for classifying image inputs. For the logistic regression and support vector machine models, which took 512 x 512 image inputs and returned categorical outputs, we cross-validated by splitting the image set in half and set one half as train data and the other as test data to prevent over fitting and a high variance model. The convolutional neural network and image segmentation methods implements multiple layers, which are expanded upon below, that prevent over fitting within the models.

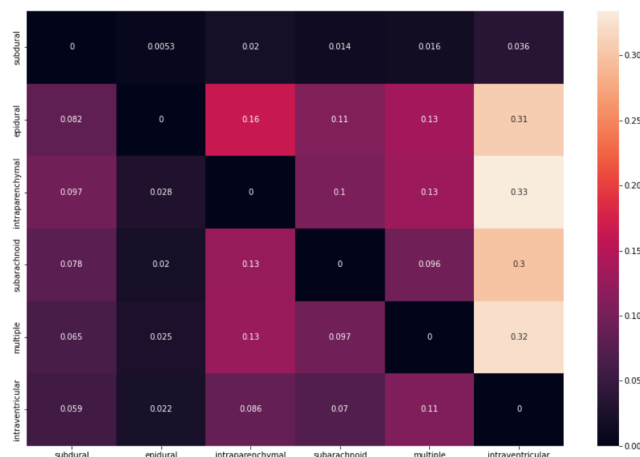
2.2 Result

2.2.1 Logistic regression

Logistic regression is a statistical method used to analyze a data set in which there are one or more independent variables that determine an outcome. It is a type of regression analysis used to predict a binary outcome based on a set of independent variables. In logistic regression, the dependent variable is a binary variable that represents the outcome of interest, and the independent variables can be continuous or categorical.

We used around 60000 input image with 512 x 512 size and the data was divided into half for both testing and training for this model. Additionally, in order to save memory the input image is shrink from 512x512 to 100x100 and the data is normalized to make the iteration run faster. We initially run 40 iterators and the result yields an accuracy of 0.67556. After increasing the number of iterations up to 100, 500, 1000. We noticed that the accuracy will be increasing to near 87%.

Based on the data and model, we generated a confusion matrix is generated to evaluate the performance of a binary classification model by comparing the predicted and actual values of the target variable which are 6 different type of hemorrhages. Below is the confusion matrix of 40 iterations.



2.2.2 Convolutional Neural Network

Convolutional Neural Networks often abbreviated as CNNs are a type of deep learning algorithm designed to process input data in the form of arrays or tensors. CNNs typical consists of three different types of layers. The first type of layer is the convoultional layer, often described as the foundation of these networks. This layer applies different filter to the input data with the goal of highlighting different shapes or patterns within the input. The next type of layer is the pooling layer. The pooling layer down samples the output of the previous layer by reducing the dimensionality while also preserving the most important features of the data. Fully connected layers are the last type of layer. These layers perform regression or classification operations on the data generated by the convolutional and pooling layers. The model implemented in this study uses two convolutional layers and two pooling layers, followed by multiple processing layers.

In this model, we use Keras API and Tensorflow backend and we divided the data into 30000 input for training and 10000 input image for testing. We used 2 convolutional layers, 2 max poolings, flatten and dense. In 2D convolutional layers, it performs 2D convolution on the input image, using a specified number of filters (32, 64, and 128 in this case) and a kernel size of 3x3 pixels. The 2D maxpooling performs max pooling on the output of the convolutional layer, reducing the spatial dimensions of the output by a factor of 2. This helps to reduce the computational complexity of the model and prevent overfitting. We use flatten to faltten the output of the previous layer into a 1D vector, which can be fed into a dense layer. The second dense layer has 6 units, which corresponds to the number of classes in the classification problem. Then using the softmax function to convert the output of the previous layer into a probability distribution over the 6 classes. The model has total 6,650,438 trainable parameters.

```
Model: "sequential"
Layer (type)                Output Shape              Param #
-----
conv2d (Conv2D)              (None, 98, 98, 32)        896
max_pooling2d (MaxPooling2D) (None, 49, 49, 32)        0
conv2d_1 (Conv2D)             (None, 47, 47, 64)       18496
max_pooling2d_1 (MaxPooling2D) (None, 23, 23, 64)        0
conv2d_2 (Conv2D)             (None, 21, 21, 128)      73856
max_pooling2d_2 (MaxPooling2D) (None, 10, 10, 128)       0
flatten (Flatten)             (None, 12800)             0
dense (Dense)                 (None, 512)               6554112
dropout (Dropout)             (None, 512)               0
dense_1 (Dense)               (None, 6)                 3078
Total params: 6,650,438
Trainable params: 6,650,438
Non-trainable params: 0
```

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We use Adam optimizer to update the weights of the model during training and categorical cross-entropy loss to evaluate the performance of the model during training. After running 10 epochs, the accuracy of the model was 93.36% and the loss was 53.38%

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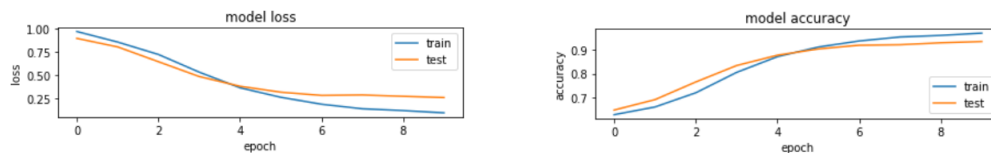


Figure 1: CNN model loss and accuracy

2.2.3 Support vector machines

Support Vector Machines (SVM) is a powerful machine learning algorithm used for classification and regression analysis. The kernel trick is a technique used to transform the data into a higher-dimensional space where it can be more easily separated. We chose a suitable kernel function; for intracranial brain hemorrhages, a radial basis function (RBF) kernel is commonly used and has shown to improve accuracy[3]. During training, the SVM algorithm will identify the optimal hyperplane that separates the data into different classes. To optimize the learning, we can adjust the hyperparameters which are parameters that are not learned during the training process, but are set prior to training. The choice of hyperparameters can have a significant impact on the performance of the SVM model and there is a large amount of freedom due to the extra features offered to us in this data set such as agreement and labeling state. For our specific data set, we performed a grid search to identify the optimal values for the hyperparameters. We then evaluated the performance of the SVM model using the accuracy and F-score metric. In summary, SVM with kernel trick can be used to classify intracranial brain hemorrhages by collecting and preprocessing the data, splitting it into training and testing sets, choosing a suitable kernel function, training the SVM model, testing the SVM model, optimizing the hyperparameters, and evaluating the performance of the model.

2.2.4 Segmentation

The process of image segmentation focuses on dividing an input image into binary classes or multiple classes depending on the segmentation model. The process of Image Segmentation is closer to that of pixel classification in comparison to image classification. The Image Segmentation model developed for this study focused on the semantic segmentation of the brain scan images to define masks for the area of brain hemorrhaging. Semantic Segmentation is the process of assigning a specific class or label to each individual pixel within the input image. For this implementation as the model focused on binary mask, the model would assign each pixel one of two labels, the first being non hemorrhage area and the second being the specific hemorrhaging area. For example using the four 512x512 windows of the brain imaging a mask would be generated that would mark the specific pixels within the 512x512 area of the hemorrhaging reference figure below:

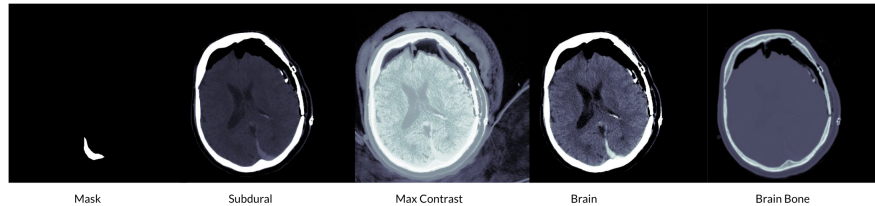


Figure 2: A 512 x 512 mask generated

Image Segmentation, specifically semantic segmentation can be an difficult task to implement due to various data parameters. Many times the area between the labels of the different classes in the output masks can be drastically different resulting in imbalanced data. In this study many of the image mask have imbalanced data where the area of the hemorrhage is much smaller than the overall area of the image. This results in model inaccuracies and the need to weight the class labels in order to maximize the value of the smaller classes when optimizing metrics and loss functions. The data was weighted towards the area of hemorrhaging in a 7:3 ratio. The model of Image Segmentation used in this study is UNet which is a convolutional neural net developed for biomedical image segmentation. The model works by following an encoder-decoder scheme that repeatedly down scales and up scales the image to generate the out image mask. The model took in input data (512, 512, 12) of the 4 windows of the brain scans and output a mask (512, 512, 2) that represented the area of brain hemorrhaging. The neural net had 11 layers each of which would scale the image up or down following the UNet model there were 5 layers for contraction the and 5 layers for expansion. The model used Dropout layers to ensure that the models was not over fitting the training data. The model had over 6,500,000 parameters.

```

=====
Total params: 6,676,610
Trainable params: 6,676,610
Non-trainable params: 0

```

```

None
(None, 512, 512, 12)
(None, 512, 512, 2)

```

The result of the model were promising. The model was trained on 2804 input made up of four images and one image mask. The data was split into training and validation sets with a ratio of 7:3. The model was trained over 50 epochs. The model tracked total loss formulated as the sum of dice loss and focal loss, as well as the metrics of IOU and Jaccard Coefficients. The model ended with a loss of 85%, an IOU value of 15%, and a Jaccard Coefficient of 99.3%.

```

Epoch 46/50
184/184 [=====] - 46s 250ms/step - loss: 0.8572 - iou_score:
0.1495 - jaccard_coef: 0.9931 - val_loss: 0.8569 - val_iou_score: 0.1495 - val_jaccard_coef: 0.9934
Epoch 47/50
184/184 [=====] - 46s 250ms/step - loss: 0.8572 - iou_score:
0.1495 - jaccard_coef: 0.9931 - val_loss: 0.8569 - val_iou_score: 0.1495 - val_jaccard_coef: 0.9934
Epoch 48/50
184/184 [=====] - 46s 251ms/step - loss: 0.8572 - iou_score:
0.1495 - jaccard_coef: 0.9931 - val_loss: 0.8569 - val_iou_score: 0.1495 - val_jaccard_coef: 0.9934
Epoch 49/50
184/184 [=====] - 46s 251ms/step - loss: 0.8572 - iou_score:
0.1495 - jaccard_coef: 0.9931 - val_loss: 0.8569 - val_iou_score: 0.1495 - val_jaccard_coef: 0.9934
Epoch 50/50
184/184 [=====] - 46s 251ms/step - loss: 0.8572 - iou_score:
0.1495 - jaccard_coef: 0.9931 - val_loss: 0.8569 - val_iou_score: 0.1495 - val_jaccard_coef: 0.9934

```

3 Conclusion

3.0.1 Logistic Regression

The model was trained with different numbers of iterations, and it was observed that the accuracy increased from 67.5% with 40 iterations to almost 87% with 1000 iterations. A confusion matrix was generated to evaluate the performance of the binary classification model for the six different types of hemorrhages. Although, the results suggest that the logistic regression model was able to accurately predict the presence of hemorrhage in the input images with a high degree of accuracy, it seems like it has the worst accuracy amongst the others.

3.0.2 CNNs

After training, the model achieves a final accuracy of 78.27% on the training set and 81.88% on the validation set. This suggests that the model is able to generalize well and not overfitting the training data. The validation accuracy is higher than the training accuracy, indicating that the model is not overfitting to the training data. However, more insights can be obtained by analyzing the confusion matrix and visualizing the training and validation curves.

3.0.3 Segmentation

While the overall implementation of the UNet model was relatively effective. The model did struggle to accurately predict the binary masks in comparison to other models implemented in this study. This is partial caused by issues with the input data as well as the relatively small data set size of 2800 cases for the model to be trained on. The overall data also did not match much of the medical image data used in industry that the UNet model was originally built to utilize. The input brain scan images were in jpg format rather than that of the more common dicom format. This led to some issues where the center of the brain images were different for each input. This is an issue as the data was imbalanced due to the large amount of black pixels outside of the brain images that could not be effectively cropped out. Even with these issues the overall implementation was effective and led to solid results for an image segmentation model. With more fine tuning, data augmentations, and more controlled input we believe the segmentation model could be significantly more effective in the future.

References

- [1] Tenny, Steven. "Intracranial Hemorrhage - StatPearls - NCBI Bookshelf." National Center for Biotechnology Information, Nation Library of Medicine.
- [2] "U-Net: Convolutional Networks for Biomedical Image Segmentation." University of Freiburg
- [3] McDermott, Barry, et al. "Brain Haemorrhage Detection Using a SVM Classifier with Electrical Impedance Tomography Measurement Frames." PLOS ONE, edited by Ruxandra Stoean, no. 7, Public Library of Science (PLOS), July 2018, p. e0200469. Crossref, doi:10.1371/journal.pone.0200469.
- [4] Xue, Zhiyun et al. "Window classification of brain CT images in biomedical articles." AMIA ... Annual Symposium proceedings. AMIA Symposium vol. 2012 (2012): 1023-9.
- [5] A. M. Patel et al., "Intracerebral Haemorrhage Segmentation in Non-Contrast CT," Scientific Reports, vol. 9, no. 1, Nov. 2019, doi: 10.1038/s41598-019-54491-6.
- [6] Wang, Xiyue et al. "A deep learning algorithm for automatic detection and classification of acute intracranial hemorrhages in head CT scans." NeuroImage. Clinical vol. 32 (2021): 102785. doi:10.1016/j.nicl.2021.102785
- [7] Lee, J.Y., Kim, J.S., Kim, T.Y. et al. Detection and classification of intracranial haemorrhage on CT images using a novel deep-learning algorithm. Sci Rep 10, 20546 (2020). <https://doi.org/10.1038/s41598-020-77441-z>